Hubert Antony

Mechanical Engineer (M.Sc.) | Additive Manufacturing PBF-LB/M, Materials Engineering, Metallography

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Residence permit under section 20 AufenthG with permission to work. I apply for the EU Blue Card myself after signing. No employer sponsorship required. Available immediately.

PROFILE

Engineering is a continuous learning process for me, and that is how I approach parameter development in additive manufacturing. I develop and qualify parameter sets against density, surface topography and process stability, and I carry out the metallography, microscopy and statistical evaluation myself. I solve technical problems by combining hands-on machine and laboratory experience with independent problem-solving, creativity and close teamwork. I work calmly under pressure and run several test series in parallel. The results are published peer-reviewed in Procedia CIRP. Alongside my studies and research I have taught CAD and simulation freelance since 2024, which keeps my Creo, CATIA, ANSYS and GD&T work current.

SKILLS

Additive Manufacturing: PBF-LB/M (also called L-PBF or SLM), build preparation in Materialise Magics and Autodesk Netfabb, powder handling, support strategy, parameter sets for FeSi6.5, AlSi10Mg and cobalt-chromium

Materials Analysis: Metallography and sample preparation, optical microscopy, porosity and density measurement, surface metrology (KEYENCE VR-3100, MarSurf M400), dilatometry and TMA (NETZSCH DIL 402, METTLER TOLEDO)

Experimental Work and Evaluation: Independent planning and execution of test series, design of experiments, Taguchi method, ANOVA and regression analysis, scientific writing, reports and publications

Design and Simulation: CATIA V5, PTC Creo, Siemens NX, ANSYS Mechanical (FEA), ANSYS Fluent (CFD), 4-axis CNC machining and CAM

PROFESSIONAL EXPERIENCE

Freelance CAD and CAE Trainer, self-employed **05/2024 - present**

Remote, with trainees and clients in India, Germany and further afield

- Teach and work hands-on with PTC Creo, CATIA V5 and ANSYS, covering GD&T, body-in-white structural design, reverse engineering and design for 3D printing.
- Trainees are international and Indian engineering students and working professionals. I design the courses, run the live sessions and write the exercise material; the client network comes from my earlier role at CADD Centre Training Services in Chennai.
- Teach design intent and manufacturability alongside the software, so that trainees leave able to design parts that can actually be made.
- Family care, 11/2025 to 04/2026: relocated to India to care for my mother during a serious illness and continued the training work remotely at a reduced load.

Academic Research Assistant, Equipment and Process Development **09/2023 - 03/2024**

Hochschule Aalen (Aalen University), LaserApplication Center, Aalen, Germany

- Upgraded a PBF-LB/M machine from design through to commissioning: simulated and designed the gas flow in the process chamber, designed and manufactured the components, then built and commissioned the shielding-gas system with a heated build platform.
- Ran external projects on additively manufactured soft magnetic materials (FeSi6.5) for electric-motor stators. Carried out the experiments and the evaluation independently; results published in Procedia CIRP (2024).
- Carried out metallographic analysis and microscopy of metal 3D-printed parts in house: sample preparation, optical microscopy and surface metrology.

Working Student, Manufacturing Engineering **05/2023 - 08/2023**

Hochschule Aalen (Aalen University), LaserApplication Center, Aalen, Germany

- Designed a shielding-gas circulation system in CATIA V5 and confirmed the design by CFD analysis in ANSYS Fluent. Manufacture and commissioning followed in the subsequent research assistant role.
- Operated metal additive manufacturing systems end to end: build preparation, powder handling, machine operation, post-processing and inspection.
- Reduced failed builds through an improved support strategy and better parameter selection.

Application Engineer, CAD and CAE **03/2022 - 09/2022**

CADD Centre Training Services Pvt. Ltd., Chennai, India

- Trained more than 100 engineers and students in design and simulation software, and wrote the course material myself.
- Taught the reverse engineering course: measuring physical parts, rebuilding them as parametric CAD models and checking the model back against the original.

• Designed automotive body-in-white structures and electric-vehicle components, and advised clients on whether a design could actually be manufactured.

CAD and CAM Engineer, CNC Manufacturing **06/2021 - 02/2022**

Axis Automation Pvt. Ltd., Chennai, India

• Programmed and operated 4-axis machining centres for precision components.

• Reverse engineered existing parts and optimised toolpaths to raise output and cut material waste.

EDUCATION

M.Sc. Advanced Materials and Manufacturing **10/2022 - 09/2025**

Hochschule Aalen (Aalen University), Aalen, Germany

• Master's thesis: optimisation of the layer topography of soft magnetic FeSi6.5 by in-process laser remelting in PBF-LB/M. Build parameters at 0.152% porosity; single remelting cut Sa from 4.55 to 2.81 µm (38.3%), every-layer cyclic remelting reached Sa 1.77 µm (more than 41% below its reference). Supervisors: Prof. Dr. Harald Riegel, Prof. Dr. Dagmar Goll.

B.E. Mechanical Engineering **07/2017 - 08/2021**

Jeppiaar Institute of Technology, Chennai, India

• Final thesis: reducing porosity in additively manufactured cobalt-chromium for medical implants, using a Taguchi design of experiments.

PROJECTS

• Thermal expansion of additively manufactured AlSi10Mg: designed a 30-geometry specimen matrix, heat treatment at 300 °C, expansion measured to 400 °C by dilatometry and TMA. Supervisor: Prof. Dr. Markus Merkel, 2023/24.

• Camera mount for a laser processing head: additively manufactured to give repeatable positioning for in-process monitoring.

PUBLICATION

Hofele M., Antony H., Kolb D., Riegel H. (2024). Production of soft magnetic FeSi6.5 parts with precise layer topography using integrated laser remelting at PBF-LB/M to induce multi-material lamination. Procedia CIRP, Volume 124, pages 6 to 11. DOI: 10.1016/j.procir.2024.08.060

LANGUAGES

• German: B1.

• English: C1, the language of my studies, my work and my publication.

• Tamil: Native speaker.

CERTIFICATIONS

• CAE and CFD using ANSYS. CIPET, Chennai, 2019

• CAD/CAM using CATIA V5 and PTC Creo. CIPET, Chennai, 2019

• Applied Ergonomics. NPTEL, IIT Kanpur, 01/2020 - 04/2020

VOLUNTEERING

• Volunteer, German Red Cross. CONTRIBUTE programme, Aalen University, 2022/23.

• Volunteer, Youth Red Cross, 2017 to 2021, and Student Coordinator of the TECH AGRONA 2K20 national project exposition, both Jeppiaar Institute of Technology, Chennai.